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SUBJECT NAME – INTRODUCTION TO WEB PROGRAMMING

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SEMESTER – 1

Q1.

HTML (Hypertext Markup Language) serves as the primary coding language for constructing web pages, providing the foundational framework for a website while working with CSS (Cascading Style Sheet) or JavaScript.

HTML's defining features include:

- Promotes ease-of-use and rapid learning
- Platform independence and compatibility via various browsers
- Use of tags to aid in content structure
- Ability to use hyperlinks; image files; tables; forms; and media files (gif's, videos, etc.)
- Ability to combine with CSS and/or JavaScript to create interactive web pages

Ways HTML is used or applied include:

- Creating web sites/web pages
- Displaying text/image/video/audio
- Creating user input forms
- Creating links between different web pages

HTML Versions Include:

- HTML 1.0- The original HTML (lightweight) supporting only basic text
- HTML 2.0 - Enhanced version supporting small amounts of text with some functionality added
- HTML 3.2 - Enhanced version with more robust functionality
- HTML 4.01 - Enhanced version adding styling and media features
- XHTML - More restrictive version of standard HTML
- HTML5 - Current version supporting audio, video, graphic images, and mobile-friendly features.

Basic Structure of an HTML Document

Every HTML document follows a standard structure.

```
<!DOCTYPE html>
<html>
<head>
  <title>My Web Page</title>
</head>
<body>
  <h1>Welcome to    My    Website</h1>
  <p>This      is      a      Sample webpage. </p>
</body>
</html>
```

Head Section

The <head> section contains information about the webpage that is not directly displayed on the screen.

It includes:

- Page title (<title>)
- Meta tags
- Links to CSS files
- JavaScript

files Example:

```
<head>
  <title>College Website</title>
</head>
```

Body Section

The <body> section contains all the visible content of the webpage.

It includes:

- Headings
- Paragraphs
- Images
- Tables
- Forms
- Links

Example

```
<body>
  <h1>Hello Students</h1>
  <p>Welcome to HTML. < /p>
</body>
```

Developing HTML for the Modern Day Web

HTML has changed greatly since its inception and has evolved to meet the needs of modern web developers. The original version of HTML was developed primarily to display text and link to one another. As the Internet has developed and improved, new features have been added.

The most significant change to HTML occurred with the rollout of HTML5, which introduced many new features, including:

- Natively supporting audio and video playback
- Improved form controls
- The ability to render graphics using the “canvas” element
- Improved performance for web pages with multimedia content

Mobile Devices and Mobile Support

HTML5 also supports mobile devices via the viewport meta tag, which will resize a web page according to the type of device being used.

Responsive Design Support

HTML allows you to create responsive design websites through both CSS media queries and flexible layout designs. With responsive design, a web page will adjust to fit the screen size of any desktop, tablet or smartphone device, providing users with a better overall experience.

Q2.

Definition of CSS

CSS (**Cascading Style Sheets**) is a style sheet language used to control the appearance and layout of HTML webpages. It helps make websites attractive and user-friendly.

Types of CSS

1. Inline CSS

- `<p style="color:blue;">Hello</p>`
- Style is applied directly to an HTML element.

2. Internal CSS

```
<style>
p{color:blue;
}
</style>
```

Style is written inside the <head> section.

3. External CSS

```
<link rel="stylesheet" href="style.css">
```

Styles are stored in a separate CSS file and linked to the webpage.

Role of CSS in Web Development:

CSS has a significant effect on the design of web pages by improving attributes like design and layout, maintaining consistent designs across multiple pages, reducing repetitively coded style properties, making websites cross/device compatible (phones/tablets/desktop), and enhancing user experience by providing appropriate styles to users.

Classes, IDs, Selectors, and Common CSS Problems:

Classes:

Classes allow you to apply the same style(s) to many elements.

```
<div class="hello"> Hello </div>
```

Id:

Id allows you to style a single element.

```
<div id="welcome">Welcome</div>
```

Selectors:

Selectors allow you to identify the appropriate html element to be styled.

Examples: `p{color: red;}` `.text{color:blue;}` `#heading{color: green;}`

Q3.

HTML Tables and Their Uses

HTML tables are used to organize and display data in rows and columns.

Common table tags:

- `<table>` – Creates a table
- `<tr>` – Creates a row
- `<th>` – Creates a heading cell
- `<td>` – Creates a data

cell Example:

```
<table border="1">
<tr>
<th>Name</th>
<th>Age</th>
</tr>
<tr>
<td>Rahul</td>
<td>20</td>
</tr>
</table>
```

Tables help present information in a clear and organized manner, making data easier to read and compare.

Types of Lists

1. Ordered List ()

- Displays items with numbers.

```
<ol>
<li>HTML</li>
<li>CSS</li>
</ol>
```

2. Unordered List ()

- Displays items with bullets.

```
<ul>
<li>HTML</li>
<li>CSS</li>
</ul>
```

3. Description List (<dl>)

- Used for terms and definitions.

```
<dl>
<dt>HTML</dt>
<dd>HyperText Markup Language</dd>
</dl>
```

Text Formatting Tags

- – Bold text
- <i> – Italic text
- <u> – Underlined text
- – Important text
- – Emphasized text

These tags improve readability and make important content stand out.

Accessibility and Semantic Best Practices

To ensure proper accessibility and semantic usage:

- Use <th> for table headings instead of normal cells.
- Add meaningful headings and labels.
- Use tables only for data, not for page layout.
- Use , , and <dl> according to the type of content.
- Keep list items inside tags.
- Use semantic tags like and instead of only visual tags.
- Maintain a clear structure so screen readers can understand the content.

Q4.

Structure of HTML Forms, Input Types, and Attributes

HTML forms are used to collect information from users and send it to a server for processing.

Basic form structure:

```
<form action="submit.php" method="post">
<label>Name:</label>
<input type="text" name="username">
<input type="submit" value="Submit">
</form>
```

Common Input Types

- text – Text input
- password – Password field
- email – Email input
- radio – Single choice option
- checkbox – Multiple choices
- submit – Submit button

Important Attributes

- name – Identifies the input field
- value – Default value
- placeholder – Displays hint text
- required – Makes field mandatory
- maxlength – Sets character limit

1. Interactivity in Web Applications

Forms make websites interactive by allowing users to:

- Enter personal information.
- Register and log in.
- Submit feedback and surveys.
- Search for information.
- Upload files and send requests.

Without forms, users would only be able to view content and not interact with the website.

2. Form Validation, User Experience, and Accessibility

Form Validation Techniques

Form validation ensures that users enter correct data.

- Use required to prevent empty fields.
- Use type="email" for valid email addresses.
- Use maxlength to limit input length.
- Use JavaScript for advanced validation.

Enhancing User Experience

- Provide clear labels and instructions.
- Use placeholders for guidance.
- Display error messages clearly.
- Group related fields together.
- Keep forms simple and easy to fill.

Improving Accessibility

- Use <label> tags for all input fields.
- Ensure proper keyboard navigation.
- Use meaningful field names.
- Provide clear error messages.
- Maintain good color contrast and readable text.

Q5.

1. Adding and Aligning Text in HTML

Text can be added in HTML using different tags.

Common Text Tags

- <h1> to <h6> – Headings
- <p> – Paragraph
- – Bold text
- <i> – Italic text
- <u> – Underlined

text Example:

```
<h1>Welcome</h1>
<p>This is a paragraph.</p>
```

Text Alignment

Text alignment can be done using CSS.

```
<p style="text-align:center;">Centered Text</p>
```

Types of alignment:

- Left
- Right
- Center
- Justify

Proper text formatting improves readability and presentation.

2. Adding Images and Hyperlinks in

HTML Images

Images are added using the tag.

```

```

Important attributes:

- src – Image location
- alt – Alternative text
- width and height – Image size

Hyperlinks

Hyperlinks are created using the <a> tag.

```
<a href="https://example.com">Visit Website</a>
```

The href attribute specifies the destination of the link.

3. Image Optimization and Accessible Links

Best Practices for Image Optimization

- Use compressed images to reduce loading time.
- Choose suitable formats such as JPG, PNG, or WebP.
- Specify image dimensions.
- Use meaningful file names.

- Add alt text for accessibility.

Best Practices for Hyperlinks

- Use clear and descriptive link text.
- Avoid vague text like "Click Here".
- Ensure links work correctly.
- Make links easy to identify.
- Use relevant keywords for better SEO.

Accessibility and SEO Benefits

- Alt text helps screen readers understand images.
- Descriptive links improve navigation for users.
- Search engines can better understand webpage content.

Q6.

1. JavaScript Syntax Rules, Data Types, and Operators

JavaScript is a scripting language used to make web pages interactive.

Basic Syntax Rules

- Statements end with a semicolon (;).
- Variables are declared using var, let, or const.
- JavaScript is case-sensitive.
- Code is written inside <script> tags.

Common Data Types

- **Number** – 10, 25.5
- **String** – "Hello"
- **Boolean** – true, false
- **Array** – [1, 2, 3]
- **Object** – {name:"Rahul"}

Common Operators

- Arithmetic: +, -, *, /
- Comparison: ==, !=, >, <
- Logical: &&, ||, !

2. Concept of JSON

JSON (**JavaScript Object Notation**) is a lightweight format used to store and exchange data between applications.

Example:

```
{  
  "name":"Rahul","age":20  
}
```

Features of JSON:

- Easy to read and write.
- Lightweight and fast.
- Language-independent.
- Widely used in web applications.

JSON represents data as key-value pairs.

3. JSON Parsing, String Conversion, and

Applications Converting JSON String to Object

JSON.parse() is used to convert a JSON string into a JavaScript object.

```
let obj = JSON.parse('{"name":"Rahul"}');
```

Converting Object to JSON String

JSON.stringify() is used to convert a JavaScript object into a JSON string.

```
let str = JSON.stringify({name:"Rahul"});
```

Applications of JSON

- Data exchange between client and server.
- Web APIs and REST services.
- Storing configuration data.
- Sending and receiving data in AJAX requests.
- Communication between web and mobile applications.